Review: The real number system

1 Consider the folllowing sets of numbers.

- $\{0, 1, 2, 3, \ldots\}$
- $\{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$
- $\{1, 2, 3, \ldots\}$
- $\{\ldots, -3, -2, -1, 0\}$

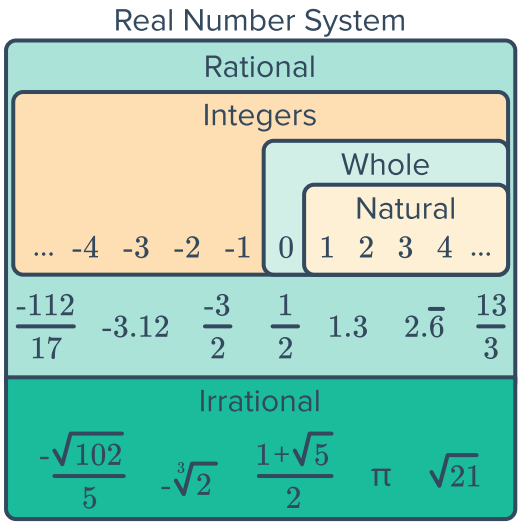
a Identify the set of all integers.

b Identify the set of whole numbers.

c Identify the smallest whole number.

2 Classify the set of numbers containing $\{0, 1, 2, \ldots\}$

3 Consider the following diagram:



a Determine whether each of the following statements is true or false.

i $\sqrt{13}$ is a rational number.

ii When written in decimal form, irrational numbers have a repeating pattern of decimal digits.

iii A real number is either a rational or irrational number.

b Classify the following numbers:

i 38 **ii** -42 **iii** $\sqrt{6}$ **iv** 3.058

v $\sqrt{49}$ **vi** $\frac{\pi}{3}$ **vii** $0.5777\ldots$ **viii** $\frac{26}{45}$

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4 Every whole number is also a rational number but not every rational number is a whole number. Explain why this statement is true.

5 Christa decides to express her favorite rational number as a decimal. Describe this decimal.

6 Classify the set of number most appropriate to describe the following:

- a The populations of towns
- b The times of the runners (in seconds) in a $100\ \mathrm{m}$ sprint.
- c Distance between Jupiter and Venus.
- d The position of a submarine relative to sea level. (Note that: height above sea level is expressed as a positive quantity.)
- e The goal difference of a hockey team at the end of a season. (Note that: The goal difference is the number of goals the team scored in the season minus the number of goals that were scored against them.)
- f The cost (in dollars) of sending a text message.

7 The circumference of a circle is given by $2\pi r$, where r is the radius.
State whether each of the following statements is true about the circumference of the circle.

- a The circumference is an irrational number if the radius is rational.
- b The circumference is a rational number if the radius is equal to π.
- c The circumference is a rational number if the radius is irrational.
- d The circumference is a rational number if the radius is equal to $\frac{a}{\pi}$ where a is an integer.

Additional questions

8 Give a definition for the subset of rational numbers.

9 Give a definition for the subset of irrational numbers.

10 Can a number be both rational and irrational? Explain.

11 Do integers belong to the subset of rational or irrational numbers? Explain.

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12 The following numbers are given in their simplest form.
For each number, state whether it is rational or irrational.

a $-\frac{2}{7}$ **b** $\sqrt{2}$ **c** 8 **d** 20.57

e $3\sqrt{5}$ **f** $1-\sqrt{7}$ **g** $-0.\overline{3}$ **h** π

13 The following numbers are not given in their simplest form.
Simplify each number and state whether it is rational or irrational.

a $\sqrt{16}$ **b** $\frac{2\pi}{8}$ **c** $-\frac{4\pi}{5\pi}$ **d** $\frac{0.4}{5}$

14 Akin states that $24.737337333...$ is a rational number because it has a repeating pattern.
Is Akin correct? Explain.

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